

Fan Qiao

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EDUCATION

Peking University, Beijing, China

Sep 2025 – Present

Ph.D. student; research focus in Control Science
School of Advanced Manufacturing and Robotics

Beihang University, Beijing, China

2021 – 2025

Undergraduate studies in Automation

GPA: 3.85/4.00 **Rank:** 1/154

RESEARCH INTERESTS

Robotic manipulation and skill transfer, with a focus on surgical robotics.

PUBLICATIONS

Propulsion Control of Bionic Robotic Fish Based on Deep Deterministic Policy Gradient Algorithm

Dong Xu, Heyang Feng, **Fan Qiao**, Kaiyang Lu, Xiaoguang Hu, and Yupeng Liu.

IEEE Open Journal of the Industrial Electronics Society, vol. 6, pp. 1496–1507, 2025. **JCR Q1 (2024)**.

RESEARCH EXPERIENCE

AutoSurge: Autonomous Surgical Robot Manipulation

Ongoing

Peking University

Manuscript in preparation

- Contributed extensively across the development and evaluation of a multi-arm system for long-horizon autonomous surgical manipulation.
- Participated in integrating safety-aware hierarchical planning, respiratory-phase-conditioned vision-language-action policies, and closed-loop anomaly management.
- Contributed to in vivo evaluation and manuscript revision. The system completed a predefined workflow of cholecystectomy steps in three live pigs, with 75/75 correct workflow transitions.

Training Deep Neural Networks

Jan 2024 – May 2024

LeapLab, Tsinghua University

Worked with Prof. Gao Huang and Yulin Wang

- Designed and implemented experiments on DeiT models with different width-to-depth ratios to investigate training difficulties in deeper networks.
- Analyzed layer-wise gradients and loss landscapes, and explored learning-rate decay strategies.

Deep Reinforcement Learning for Efficient Robotic Fish Propulsion

2023 – 2024

School of Automation Science and Electrical Engineering, Beihang University

Worked with Assoc. Prof. Dong Xu

- Implemented DDPG-based dynamic joint stiffness control for a two-joint robotic fish and contributed to manuscript writing.
- The study demonstrated approximately 9.77% higher system efficiency through simulation and physical experiments; published in *IEEE OJIES* (2025).

SELECTED HONORS AND AWARDS

National Scholarship (2 times, Top 2%)

2023, 2024

Gold Medal, 51st International Exhibition of Inventions Geneva

2026

Team award for an AI-enhanced surgical teleoperation system.

First Prize, Beijing College Mathematics Competition

2021, 2022

SKILLS

Programming and tools: C/C++, Python, L^AT_EX. **English:** CET-6, 598.